INCH-POUND

MIL-PRF-32717
21 August 2022

PERFORMANCE SPECIFICATION

CLOTH, KNITTED, SILK WEIGHT, STRETCH, FLAME RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types and classes of multi-directional stretch, flame-resistant, moisture management silk weight cloth used in the manufacture of long underwear shirts and drawers.

1.2 Classification.

1.2.1 Types. The silk weight fabric is furnished in the following types as specified (see 6.2):

Type I - Moderate stretch
Type II - Significant stretch

1.2.2 Classes. The silk weight fabric is furnished in the following classes as specified (see 6.2):

Class 1 - Black 3241
Class 2 - Coyote Brown 3758

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to

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ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-1491 Glossary of Knitting Imperfections

(Copies of this document are available at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at http://www.ftc.gov.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

ENVIRONMENTAL PROTECTION AGENCY EPA chemicals-and-toxics-topics Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)

(Copies are available online at https://www.epa.gov/environmental-topics/chemicals-and-toxics-topics.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1	Gray Scale for Color Change
AATCC Evaluation Procedure 2	Gray Scale for Staining
AATCC Test Method 8	Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 15	Colorfastness to Perspiration
AATCC Test Method 16.3	Colorfastness to Light: Xenon Arc
AATCC Test Method 20A	Fiber Analysis: Quantitative
AATCC Test Method 61	Colorfastness to Laundering: Accelerated
AATCC Test Method 79	Absorbency of Bleached Textiles
AATCC Test Method 135	Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://www.aatcc.org.)

ASTM INTERNATIONAL

ASTM D1776/D1776M	Standard Practice for Conditioning and Testing Textiles
ASTM D1777	Standard Test Method for Thickness of Textile Materials
ASTM D2594/D2594M	Standard Test Method for Stretch Properties of Knitted Fabrics Having Low Power
ASTM D3512/D3512M	Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester
ASTM D3776/D3776M	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3786/D3786M	Standard Test Method for Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method
ASTM D6413/D6413M	Standard Test Method for Flame Resistance of Textiles (Vertical Test)
ASTM E96/E96M	Standard Test Methods for Water Vapor Transmission of Materials

(Copies of these documents are available online at http://www.astm.org/.)

SDL ATLAS

Part Number 402985 – Slub/Knot Replica Set

(Replica sets are available for purchase from SDL Atlas Customer Service, see 6.4.)

INFORMA HEALTHCARE

Repeat Insult Patch Test -Modified Draize Procedure – Principles and Methods of Toxicology, (fourth edition), A Wallace Hayes (editor), pp 1057 – 1060, 2001

(Copies are available online at http://www.taylorandfrancis/.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3).

3.2 Standard sample. The cloth shall match the standard sample for shade, colorfastness and appearance on the face side and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 4.6.2).

3.3 Cloth. The cloth shall be a multi-directional, stretch, circular knitted, silk-weight fabric with spandex. The Type I cloth fiber content shall be 89 percent meta-aramid, 5 percent para-aramid, 3 percent P140 (carbon based antistatic) and 3 percent spandex. The Type II cloth fiber content shall be 82 percent modacrylic, 14 percent viscose, and 4 percent spandex blend. The color shall be as specified by the contract (see 6.2). The cloth shall meet the requirements specified in Table I when tested as specified in 4.6. The cloth shall be available in Class 1 Black 3241 or Class 2, Coyote Brown 3758.

3.4 Width. For Government procurements only, unless otherwise specified, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.5 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished full width in continuous lengths, each not less than 40 yards.

3.6 Face identification. The face side of the solid shade dyed cloth shall be identified by stamping that side with the word “FACE” at each end of the roll.

3.7 Roll identification. Each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.”

3.8 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.6, Table II. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used (see 4.7 and 6.2).

3.9 Workmanship. The finished cloth shall be uniform in quality and free from irregular defects that can adversely affect usage or durability and the defects specified in 4.4.1.2.

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TABLE I. Physical requirements.

CHARACTERISTICS	TYPE I REQUIREMENTS	TYPE II REQUIREMENTS
Weight, oz /yd^2	5.25 - 6.25	5.3 - 6.3
Dimensional Stability, max		
Wales	10%	10%
Courses	8%	5%
Bursting strength, psi, min	70	48
Pilling, min	3	3
Colorfastness to Laundering, min		
Change:	4	4
Staining:		
Acetate	4	3
Cotton	4	3.5
Nylon	3	3
Polyester	4	3.5
Acrylic	4	3.5
Wool	4	3.5
Colorfastness to Perspiration, acid, min		
Change:	4	4
Staining:		
Acetate	4	3
Cotton	4	3.5
Nylon	3	3
Polyester	4	3.5
Acrylic	4	4
Wool	4	3.5
Absorbency, seconds, max		
Face	5	5
Back	5	5
Colorfastness to Crocking (min):		
Wet	4	3
Dry	4	4
Colorfastness to Light, (170 kJ@420nm or after 40 AFU)(min):	4	4
Flame Resistance –		
After flame – wales, seconds, max	2.5	≤1.0
After flame – courses, seconds, max	2.5	≤1.0
After glow – wales, seconds, max	25.0	20.0
After glow – courses, seconds, max	25.0	20.0
Char length – wales, inches, max	5.0	6.0
Char length – courses, inches , max	5.0	6.0

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TABLE I. Physical requirements - Continued.

CHARACTERISTIC	TYPE I REQUIREMENTS	TYPE II REQUIREMENTS
Stretch & Growth		
Growth, Loose Fitting:		
After 2 hr. ±5 min:		
a. 60 sec after releasing, wale, max	5%	8%
b. 60 sec after releasing, course, max	7.0%	15%
c. 1 hr after releasing, wale, max	5.0%	5%
d. 1 hr after releasing, course, max	10.0%	12%
Stretch, Form Fitting (Type I), Loose Fitting (Type II): min		
Wales	20.0%	10%
Courses	175.0%	100%
Thickness, at 0.6 psi, inches, max	0.04	0.03
Water Vapor Transmission Rate, $g/m^2/24$ hr, min	1350	1350

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the examinations and tests specified in 4.4, 4.5, 4.6 and 4.7.

4.2.1 First article samples. Unless otherwise specified in the contract (see 6.2), the number of samples for first article inspection shall be 5 yards. The sample unit shall be one yard and the lot size shall be expressed in units of yards.

4.3 Conformance inspection. Conformance testing shall consist of all examinations and tests specified in 4.4, 4.5, 4.6 and 4.7.

4.3.1 Conformance inspection sampling. Unless otherwise specified, sampling for inspection shall be performed in accordance with the demerit point system listed in 4.4.1. The sample unit shall be one yard and the lot size shall be expressed in units of yards.

4.3.2 In-process inspection. Visual and dimensional examinations of the fabric and its components or optional items shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with requirements specified in Section 3. Materials and components, which can be classified, as a defect in accordance with 4.4.1 through 4.4.1.2 shall be removed from production.

4.4 Inspection methods.

4.4.1 Visual examination. The finished cloth shall be examined on face side only for all defects defined in MIL-STD-1491 which are clearly visible at normal inspection distance (3 feet). It shall be scored and assigned demerit points except that only those slubs and knots which exceed the limits show SDL ATLAS for slubs and “C” for knots, shall be scored. No linear yard (increment of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 45 points. If one roll exceeds the point level per 100 square yards, a second sample of 20 rolls shall be examined only for the individual roll quality. If one or more rolls in the second sample of 20 exceeds the individual point value, the lot shall be rejected. Point computation for the lot quality and individual roll quality shall be as follows:

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Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to 22,000	3
22,001 and over	5

4.4.1.1 Demerit point computation.

$$\text{Points per 100 square yards} = \frac{3600\ T}{(W \times Y)}$$

where:
T = total number of points scored in the sample
W = width (in inches) of the cloth purchased
Y = total number of yards inspected

4.4.1.2 Demerit points.

a. For defects up to and including 3 inches in any dimension – one point.
b. For defects exceeding 3 inches but not exceeding 6 inches in any dimension – two points
c. For defects exceeding 6 inches but not exceeding 9 inches in any dimension – three points
d. For defects exceeding 9 inches in any dimension – four points.

The following defects, when present, shall be scored four points for each yard in which they occur.

a. Baggy, ridgy or way cloth. Edges are rolled, folded, doubled, scalloped or wavy.
b. Any color off shade, not uniform, (poor dye penetration, mottled, spotty, streaky or cloudy (face side only).
c. Excessive neppiness
d. Any hole, cut or tear.
e. Run or drop stitch
f. Missing yarn or missing knit.
g. Crease or wrinkle embedded.
h. Any solid lump, defined as a slub C or 4, or knot that exceeds level C on the respective SDL ATLAS scale.
i. Spot, stain or foreign matter 1/
j. Gross length 2 or more yards less than the ticketed length.

1/ Clearly visible at normal inspection distance (approximately 3 feet).

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4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off-shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot presented by the sample (see 6.2).

4.4.3 Face identification and nonconformance with the textile identification act examination. During the yard-by-yard examination, each roll in the sample shall be examined for nonconformance to the face identification and Textile Fiber Products Identification Act. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, two or more rolls in the sample are not labeled or ticketed in accordance with the Textile Fibers Products Identification Act (see 6.2).

4.4.4 Examination for length. Rolls shall be examined for gross length. When total yardage in the roll exceeds 100 yards, only 100 yards shall be examined (see 6.2).

4.5 End item testing. The cloth shall be tested for characteristics listed in Table II. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 5 continuous yards, full width of finished cloth. The sample size shall be as follows and the lot shall be unacceptable if one (1) or more sample units fail to meet any requirement.

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to 22,000	3
22,001 and over	5

4.5.1 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the samples is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls (see 6.2).

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4.6 Methods of inspection.

TABLE II. End item tests.

Characteristic	Requirement Paragraph	Verification methods
Visual Shade Matching	3.3	4.6.1
Weight	3.3	ASTM D3776/D3776M, (Option C)
Colorfastness to Laundering	3.3	AATCC Test Method 61 – Opt 2a, 3 cycles, 1/, 2/
Colorfastness to Light: Xenon	3.3	AATCC Test Method 16.3, Option 3 1/
Colorfastness to Crocking	3.3	AATCC Test Method 8, 2/
Dimensional Stability	3.3	AATCC Test Method 135, 1, III, A, ii, 3 cycles
Thickness	3.3	ASTM D1777
Growth	3.3	ASTM D2594/D2594M, Loose Fit
Stretch	3.3	ASTM D2594/D2594M, Loose Fit (Type I), Loose Fit (Type II)
Flame Resistance (Vertical)	3.3	ASTM D6413/D6413M
Colorfastness to Perspiration	3.3	AATCC Test Method 15, 1/, 2/
Pilling	3.3	ASTM D3512/D3512M
Bursting Strength	3.3	ASTM D3786/D3786M
Absorbency	3.3	AATCC Test Method 79
Water Vapor Transmission Rate, Procedure B	3.3	ASTM E96/E96M, 3/
Fiber Identification	3.3	AATCC Test Method 20A
Toxicity	3.8	4.7

1/ The specimen shall be rated using AATCC Evaluation Procedure 1.
2/ The specimen shall be rated using AATCC Evaluation Procedure 8.
3/ The test shall be conducted with a 550 ft/min velocity and measuring only once at 24 hours.

4.6.1 Visual shade matching. The color and appearance of the finished cloth shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option C (2017e) with sources simulating artificial daylight D65 illuminant having a correlated color temperature of 7500 (± 200) Kelvin, with illumination of 100 (±20) foot candles shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856 (±200) K (see 3.2.)

4.7 Toxicity test. Unless otherwise specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the finished cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.8) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the fabrication of multi-climate protective clothing.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Applicable Government including any revisions and non-government documents cited under Section 2.
c. Type, class, and quantity desired (see 1.2).
d. National stock number.
e. First article and conformance inspection requirements (see 4.2 and 4.3).
f. Name and address of the first article inspection facility; and the name and address of the Government activity responsible for conducting the first article inspection program.
g. Packaging requirements (see 5.1).
h. When toxicity testing is required or a Certificate of Compliance (CoC) may be submitted indicating the material meets the requirement (see 3.8 and 4.7).

6.3 Toxicity. Further information on chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens may be found at http://www.epa.gov/ebtpages/pollutants.html.

6.4 Information requests. For access to information such as standard shade samples of cloth etc. go to Defense Supply Center Philadelphia's website for their Specification/Pattern/Drawing Request form, https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Requests to use equivalent materials should be sent to the contracting officer for approval by the military services.

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6.5 Source for replica sets. Replica sets are available for purchase from SDL Atlas. Contact Customer Service 1-803-329-2110 and request item number is 402985 -Slub/Knot Replica Set.

6.6 Subject term (key word) listing:

Aramid
Multi-climate protection

CONCLUDING MATERIAL

Custodians:
Navy - AS
Air Force - 11

Preparing activity:
Navy - AS

(Project 8305-2021-030)

Review activities:
Army - AR, AT, AV, MI
Navy - CG1, NU
Air Force - 99
DLA - CT

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